

- 25** A source of sound of constant power P is situated in an open space. The intensity I of sound at distance r from this source is given by the equation shown.

$$I = \frac{P}{4\pi r^2}$$

How does the amplitude a of the vibrating air molecules vary with the distance r from the source?

A $a \propto \frac{1}{r}$

B $a \propto \frac{1}{r^2}$

C $a \propto r$

D $a \propto r^2$